Rushav Dash

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EDUCATION

MSc Technology Innovation (Robotics), *University of Washington* — 09/2025 – Present | Bellevue, WA

B.S. Mechanical Engineering, *University of Washington* — 09/2021 – 06/2025 | Seattle, WA

PROFESSIONAL EXPERIENCE

Global Innovation Exchange, UW, *Prototype Lab Assistant* — 12/2025 – Present | Bellevue, WA

- Assisted students with 3D printing (FDM/SLA) and PCB cutting workflows for robotics and engineering projects; designed custom tooling and fixtures to improve lab processes and part consistency.
- Beta-tested robotics lab curriculum by completing all exercises (SLAM, LiDAR obstacle avoidance, YOLO object detection, sensor fusion, waypoint navigation) and identifying hardware integration issues, documentation gaps, and workflow improvements for 70+ graduate students.
- Troubleshot mechanical assemblies, electronics integration, and fabrication challenges; helped students refine designs for manufacturability and develop assembly procedures for sensor mounting and calibration.

Zap Energy, *R&D Engineering Intern* — 06/2024 – 09/2024 | Everett, WA

- Designed precision fuel injection system with ±10-micron tolerances for fusion reactor prototypes using SolidWorks CAD with GD&T specifications, thermal expansion FEA at 200°C, and iterative design reviews with machine shops to validate manufacturing feasibility and optimize for 5-axis CNC machining
- Engineered Arduino-controlled thermal cycling system achieving ±0.5°C precision at 250°C using closed-loop PID control, solid-staterelays, and custom PCB with hardware safety interlocks; designed enclosure and thermal management system, validated uniformityacross test chamber

Zap Energy, *R&D Engineering Intern* — 07/2023 – 09/2023 | Everett, WA

- Built high-speed force testing rig (26 kN capacity) integrating custom DAQ hardware, signal conditioning circuits, K-type thermocouples, and LabVIEW control software
- Reduced component evaluation time 40% through automated Python data analysis with NumPy/Pandas for stress-strain characterization

Posner Research Group, UW, *Research Lab Assistant* — 04/2024 – 08/2024 | Seattle, WA

- Fabricated biomimetic artificial finger with integrated strain gauge arrays for robotic tactile sensing using multi-layer Dragon Skin silicone molding with 6-micron layer precision; developed cleanroom casting procedures with controlled cure times and mold release protocols.
- Achieved 2-micron mechanical repeatability and 0.1N force resolution through iterative FEA optimization, material characterization including tensile testing and hysteresis analysis, and calibrated signal processing algorithms for strain-to-force conversion

Integrated Fabrication Laboratory, UW, *Research Lab Assistant* — 02/2023 – 06/2025 | Seattle, WA

- Built automated analysis pipeline in MATLAB and Python for characterizing 3D-printed lithium-ion batteries, streamlining print condition optimization
- Enhanced research efficiency through advanced data visualization and statistical modeling of 3D battery performance data
- Applied statistical modeling and data visualization to identify performance trends across novel electrode materials, directly informing next-iteration print parameters
- Presented battery printing research at the Mary Gates Symposium to an audience of 100+ faculty and students

PROJECTS

AlienBot, *UW TECHIN 516 Robotics Final*

- Developed a ROS2-based autonomous navigation system integrating LiDAR and depth camera sensor fusion for real-time obstacle avoidance and path planning in dynamic indoor environments
- Implemented and tuned a SLAM pipeline using Nav2 with custom tf2 coordinate transforms, achieving stable localization across multi-room environments during live demonstration

VitreaClear, *UW Mechanical Engineering Capstone*

- Designed and prototyped an ultrasound acoustic radiation force system targeting vitreous eye floaters; characterized transducer beam profiles and optimized focal geometry through iterative water bath testing; patent pending, externally funded
- Built a real-time power control feedback loop regulating acoustic output intensity, ensuring consistent force delivery within target therapeutic parameters across repeated trials

Synthetic Sleep Dataset, *UW TECHIN 513 Signal Processing Final*

- Developed a generative pipeline producing physiologically plausible synthetic polysomnography (PSG) signals (EEG, EOG, EMG) validated against real-world distributions using statistical similarity metrics
- Augmented a sleep-staging classifier training set with synthetic data, improving downstream model F1-score by 11% while reducing dependency on scarce labeled clinical recordings

Adaptive Robotics Gripper, *UW Husky Robotics*

- Designed a variable-geometry adaptive gripper mechanism in SolidWorks capable of handling objects across a 10–120 mm diameter range; fabricated and iterated prototypes using FDM 3D printing with validated force output against mechanical requirements

SKILLS

Robotics and Autonomy: ROS2, SLAM, Sensor Fusion, Computer Vision (YOLO, OpenCV), LiDAR, Depth Cameras, Path Planning, Localization, PID Control, Gazebo Simulation, Autonomous Navigation

Programming: Python, C++, MATLAB, Linux, Docker, Git

ML & Signal Processing: PyTorch, TensorFlow, CNNs, Signal Processing, Transformers

Hardware: SolidWorks, Arduino, Raspberry Pi, DAQ Systems, Sensor Integration, Mechanical Prototyping